Jaber Khan Afrooz

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Summary

Computer Science and Engineering graduate with hands-on experience designing, developing, and evaluating machine learning and AI-powered applications. Experienced in Python, deep learning, NLP, information retrieval, and backend development, with practical exposure to RAG architectures, hybrid search, LLM integration, and on-device ML deployment. Demonstrated ability to build end-to-end systems, benchmark model performance, and translate research concepts into functional applications. AWS-certified in Machine Learning and Generative AI Foundations.

Education

Osmania University 2022 – 2026
Bachelor of Engineering in Computer Science and Engineering CGPA: 8.12
ISL Engineering College, Hyderabad

Telangana Board of Intermediate Education 2020 – 2022
Sri Chandra Junior College CGPA: 9.0

Board of Secondary Education, Telangana 2010 – 2020
St. Joseph's Grammar High School CGPA: 10.0

Technical Skills

Programming: Python, SQL, Kotlin

AI & Machine Learning: Machine Learning, Deep Learning, NLP, TensorFlow, Scikit-learn, NumPy, Pandas

Generative AI & Information Retrieval: RAG, LLMs, Hugging Face Transformers, Sentence Transformers, LangChain, LangGraph, BM25, FAISS, ChromaDB, Cross-Encoders

Backend & Development: FastAPI, Streamlit, REST APIs, Android Studio, TensorFlow Lite

Tools & Cloud: Git, GitHub, AWS

Projects

Hybrid Local and Cloud RAG & Chatbot | *Python, FastAPI, ChromaDB, HNSW, Qwen2.5-3B, Gemini* **Github**

- Developed a **hybrid Retrieval-Augmented Generation (RAG) and chatbot system** supporting both locally hosted **Qwen2.5-3B** and cloud-based **Gemini** models with configurable RAG and general chat modes.
- Built a document question-answering pipeline using **PDF extraction, text chunking, Sentence Transformer embeddings, ChromaDB, and HNSW similarity search** for context-aware retrieval.
- Implemented a **FastAPI backend and vanilla HTML/CSS/JavaScript frontend** with PDF indexing, Top-K retrieval control, model-provider switching, source tracking, and end-to-end latency monitoring.

BEIR SciFact Hybrid Retrieval Pipeline | *Python, FAISS, Hybrid retrieval, Cross-Encoders* **Github**

- Built a **hybrid retrieval pipeline** over **5,183 scientific documents (12,101 chunks)**, combining BM25 lexical search and dense semantic retrieval using **all-MiniLM-L6-v2, FAISS, and Reciprocal Rank Fusion (RRF)** at approximately **8 ms/query**.
- Benchmarked four retrieval configurations across all **300 SciFact test queries** using BEIR qrels and standard metrics, achieving **81.9% Recall@10 and 0.681 NDCG@10**, outperforming BM25 (75.5% / 0.632) and dense-only retrieval (78.3% / 0.645).
- Evaluated **cross-encoder reranking**, finding negligible improvement in NDCG@10 (0.682 vs. 0.681) at approximately **80× higher latency** (0.61 s vs. 8 ms); retained hybrid RRF as the default and implemented per-query failure analysis to diagnose retrieval misses.

AI-Powered Road Accident Detection System | *Python, TensorFlow/Keras, TensorFlow Lite, Kotlin, Android Studio* **Github**

- Led a team in building a smartphone-based accident detection system using a **1D-CNN** ($\sim$115K parameters) that classifies 2-second windows of 50 Hz accelerometer and gyroscope data into **normal, vehicle accident, and pedestrian accident** states, achieving **96.6% test accuracy** (per-class F1 of 0.95–0.99) on a balanced 2,988-sample test set.
- Built the on-device pipeline in Kotlin: **sensor fusion with time-indexed linear interpolation, 8-D feature extraction, Z-score normalization, and multi-frame temporal confirmation** to cut false positives, running **TensorFlow Lite** inference on a dedicated background thread inside a wake-lock-protected foreground service.
- Stress-tested the model with confusion-matrix, per-class ROC/AUC, and sensor-drift analysis, retaining **93.8% accuracy** under heavy bias drift (σ=0.4); trained with early stopping and LR scheduling.
- Implemented the emergency response flow: lockscreen alert with an adaptive **15s/30s countdown**, **fused GPS tracking, automated SMS alerts with Google Maps links**, manual SOS, and home-screen widgets for one-tap control.

AI vs. Human Text Detector | *Python*, *scikit-learn*, *XGBoost*, *PyTorch*, *Hugging Face Transformers*, *FastAPI* **Live**

- Engineered an end-to-end **ML classification pipeline** for a **161K-sample dataset**, implementing text preprocessing, deduplication, stratified 80/10/10 splits, and a **200K-dimensional hybrid TF-IDF feature pipeline** with leakage-safe training and evaluation.
- Designed and benchmarked multiple classification architectures, including **Linear SVM, XGBoost, and fine-tuned DistilRoBERTa**; achieved **98.5% and 99.7% test accuracy** with Linear SVM and DistilRoBERTa, respectively, using GPU-accelerated FP16 fine-tuning.
- Built a dedicated **out-of-distribution evaluation framework** with 190 difficulty-graded synthetic samples, uncovering substantial generalization gaps (83% Linear SVM, 59% XGBoost, 53% DistilRoBERTa); investigated dataset bias, source distribution, and genre-dependent classification patterns.
- Developed a **production-oriented inference service** using FastAPI and vanilla JavaScript, integrating model loading, probability-based predictions, confidence scoring, text diagnostics, short-input reliability warnings, and error handling; consolidated trained models and vectorizer into a reusable **Joblib artifact** with standalone evaluation tooling.

Training & Coursework

AWS Generative AI Virtual Internship 2025

- Completed applied coursework and hands-on labs covering generative AI foundations and AWS ML services.

AI & Machine Learning Virtual Internship 2025

- Applied deep learning and ML techniques to guided project work, building foundational experience ahead of independent CNN and NLP projects.

Google Android Developer Virtual Internship 2024

- Built Android application fundamentals with Kotlin and Android Studio, feeding directly into the Road Accident Detection System project.

Certifications

AWS Academy Graduate - Machine Learning Foundations

AWS Academy Graduate - Generative AI Foundations

Achievements

- Achieved 97% classification accuracy with a CNN-based smartphone accident detection system using accelerometer and gyroscope data.
- Developed and deployed an AI vs. human text classifier achieving 98% test accuracy on a 164K+ sample dataset.
- Built real-time AI-powered Android applications using TensorFlow Lite and Kotlin for on-device inference.
- Developed full-stack web applications using FastAPI, HTML/CSS/JavaScript, and machine learning models for real-time predictions.